

US Traffic Road Accident Risk Factors and Hotspot Analysis

From Exploratory Data Analysis to Machine Learning-Based Severity Prediction

Data Science Certification Project - Corizo

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Dataset: US Accidents (2016–2023), Kaggle / Sobhan Moosavi et al.

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Scikit-learn, XGBoost, SHAP

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1. Objective

This project analyzes seven years of US traffic accident data to identify the factors that influence where, when, and how severely road accidents occur. It examines temporal patterns (time of day, day of week, seasonality), environmental conditions (weather, visibility, wind), and road infrastructure (junctions, traffic signals, crossings) to understand their relationship with accident frequency and severity. The project then extends this exploratory analysis into a supervised machine learning model that predicts the probability that a given accident, if it occurs, will be severe, turning descriptive analysis into a usable risk-estimation tool, and closes with an unsupervised clustering step that groups accident hotspots into risk tiers.

2. Dataset Overview

The dataset used is the US Accidents (2016–2023) dataset, a large-scale, countrywide traffic accident dataset covering road accidents across the United States.

- Coverage: 49 states across the United States
- Time period: February 2016 – March 2023
- Records: 7,728,394 accident reports
- Format: CSV
- Features: 46 attributes covering location, time, weather, and road infrastructure
- Source: Moosavi, S., Samavatian, M. H., Parthasarathy, S., Ramnath, R.

Initial inspection showed the raw dataset consumed approximately 6 GB of memory, which motivated the data-type optimization step described in Section 3.2.

3. Methodology

3.1 Data Cleaning

Columns not useful for analysis were removed, including ID, Source, Street, Timezone, Airport_Code, Zipcode, Description, End_Lat, End_Lng, and Wind_Chill(F), either identifiers, redundant with other fields, or excessively granular.

Missing values were then handled based on variable type:

- Numerical weather variables (Temperature, Humidity, Pressure, Visibility), imputed with the median, which is robust to outliers.
- Precipitation, missing values set to 0, assuming no recorded precipitation implies dry conditions.
- Categorical variables (Wind_Speed, Wind_Direction, Weather_Condition, Sunrise_Sunset, twilight indicators), imputed with the mode.
- Weather_Timestamp, forward-filled.
- Rows with missing Start_Time, End_Time, or City, dropped, since these are essential fields that cannot be reasonably imputed.

No duplicate records were found in the dataset. After cleaning, all remaining missing-value percentages were reduced to 0%.

3.2 Data Preprocessing & Optimization

To make the dataset more efficient to work with:

- City, County, State, Weather_Condition, Wind_Direction, and the twilight/sunrise-sunset columns were converted to the category dtype.
- Binary road-feature columns (Junction, Traffic_Signal, Crossing, Railway, etc.) were converted from True/False to 0/1.
- Start_Time, End_Time, and Weather_Timestamp were converted to proper datetime types.

These optimizations reduced the dataset's memory footprint from approximately 6 GB to 1.3 GB, roughly a 29% reduction, with the cleaned dataset containing 7,728,141 rows across 36 columns at this stage.

3.3 Feature Engineering

Twelve new features were engineered from the cleaned data to support both the exploratory analysis and the modeling stage:

- Accident_Duration_Min, minutes between Start_Time and End_Time.
- Day_of_Week, day name extracted from Start_Time.
- Weekend, binary flag for Saturday/Sunday.
- Month and Year, extracted from Start_Time.
- Time_of_Day, Morning / Afternoon / Evening / Night, binned from the accident hour.
- Season, Winter / Spring / Summer / Fall, derived from Month.
- Visibility_Level, Low / Medium / High, binned from Visibility(mi).
- Wind_Level, Low / Moderate / High, binned from Wind_Speed(mph).
- Weather_Category, simplified grouping (Clear, Cloudy, Rain, Snow, Fog/Mist, Storm, etc.) of the raw weather text.
- Severe_Accident, binary target: 1 for Severity 3–4, 0 for Severity 1–2.
- Rush_Hour, binary flag for 7–9 AM and 4–6 PM.

Rows with missing Start_Time/End_Time (needed for the time-based features, ~9.6% of rows) were dropped rather than imputed, to avoid introducing incorrect temporal patterns. The final, fully-engineered dataset used for analysis and modeling contains 6,985,002 rows across 49 columns.

Progress was checkpointed at three stages of this pipeline (post-cleaning, post-optimization, and post-feature-engineering) so that analysis could resume from any stage without re-running the full pipeline on the 7.7-million-row dataset.

4. Exploratory Data Analysis

4.1 Accident Overview

Accident Severity Distribution

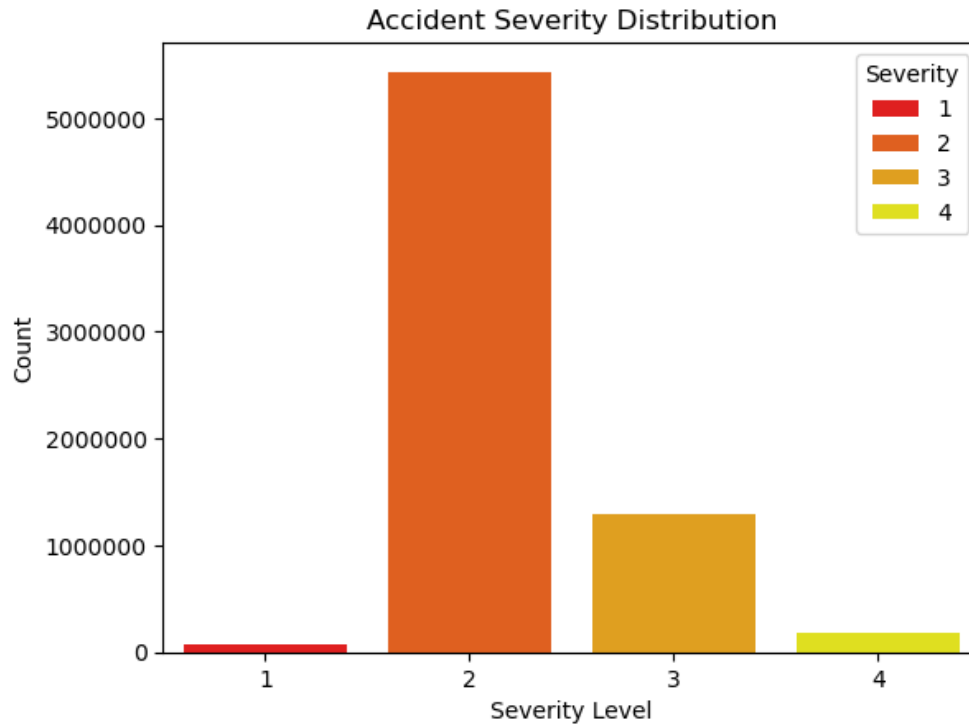


Figure 1: Distribution of accidents by severity level

Key Insights:

- The majority of accidents fall under Severity Level 2, indicating most accidents are moderately serious rather than extremely severe.
- High-severity accidents (Level 3 and Level 4) occur far less frequently, suggesting severe collisions are comparatively rare but still a meaningful safety concern.

4.2 Temporal Accident Patterns

4.2.1 Accidents by Hour of Day and Severity

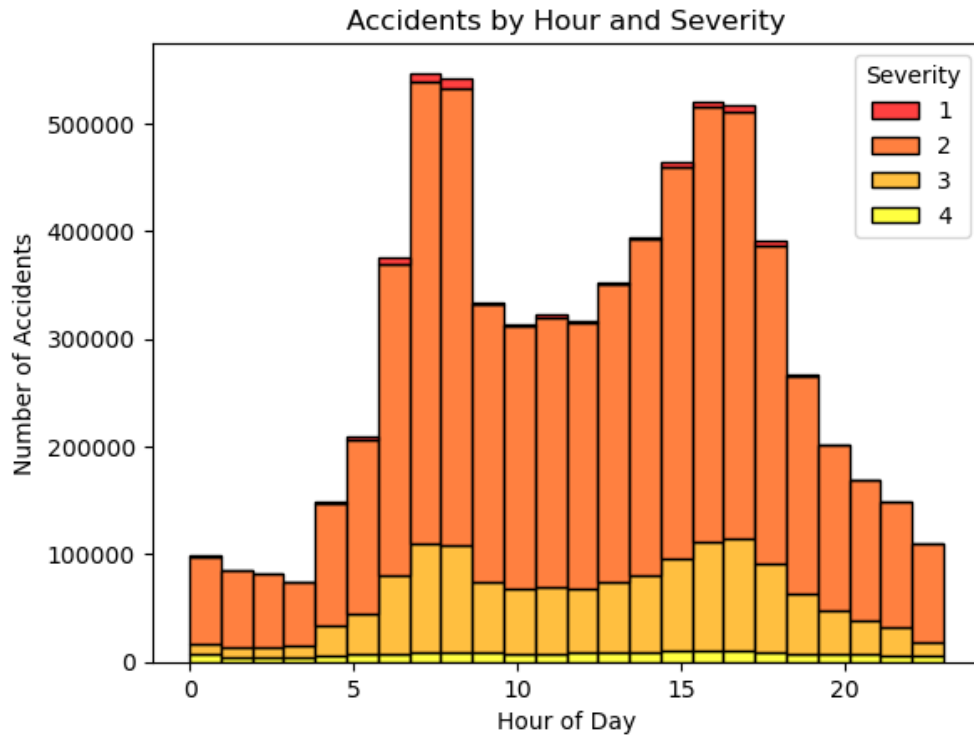


Figure 2: Accident frequency by hour of day, stacked by severity

Key Insights:

- Accidents rise sharply between 6 AM and 9 AM, peaking around 7–8 AM, the morning commute.
- A second peak occurs between 3 PM and 6 PM, mainly around 4–5 PM, corresponding to the evening commute.
- Most accidents across all hours fall under Severity Level 2.

4.2.2 Accidents by Day of Week and Rush Hour

Key Insights:

- Accident counts rise steadily from Monday to Friday, reflecting weekday commuting activity, with rush-hour accidents consistently elevated from Tuesday to Thursday.
- Friday shows the largest overall accident count, especially outside rush hours , likely reflecting increased evening/weekend travel.
- Accidents drop noticeably on Saturday and Sunday, consistent with reduced commuter traffic; rush-hour accidents in particular are much lower on weekends.

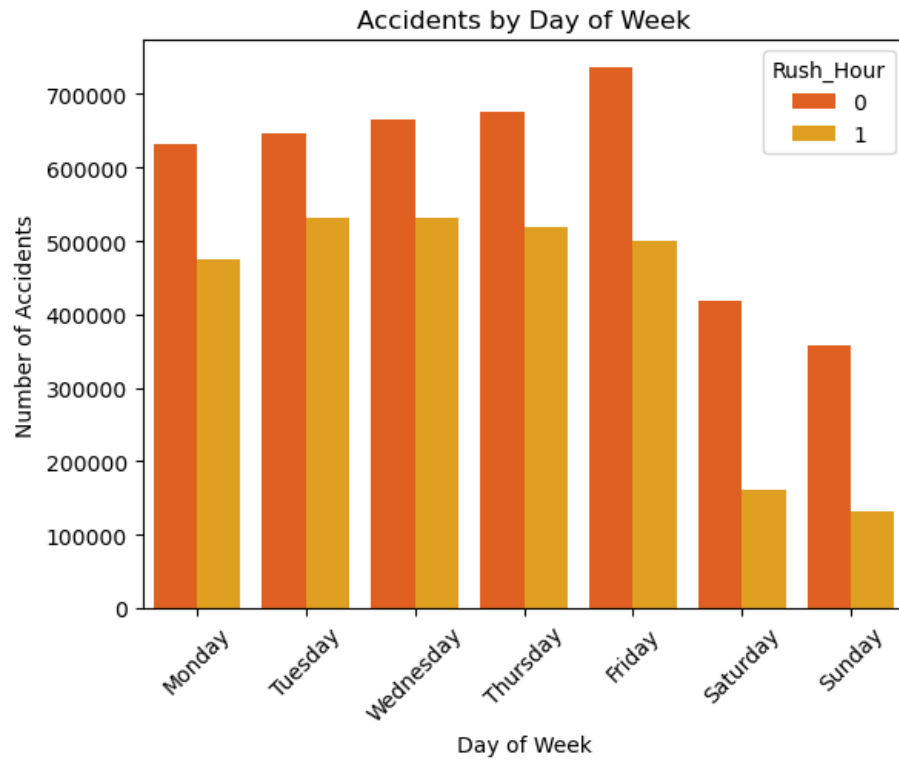


Figure 3: Accident counts by day of week, split by rush-hour status

4.2.3 Accidents by Time of Day and Rush Hour

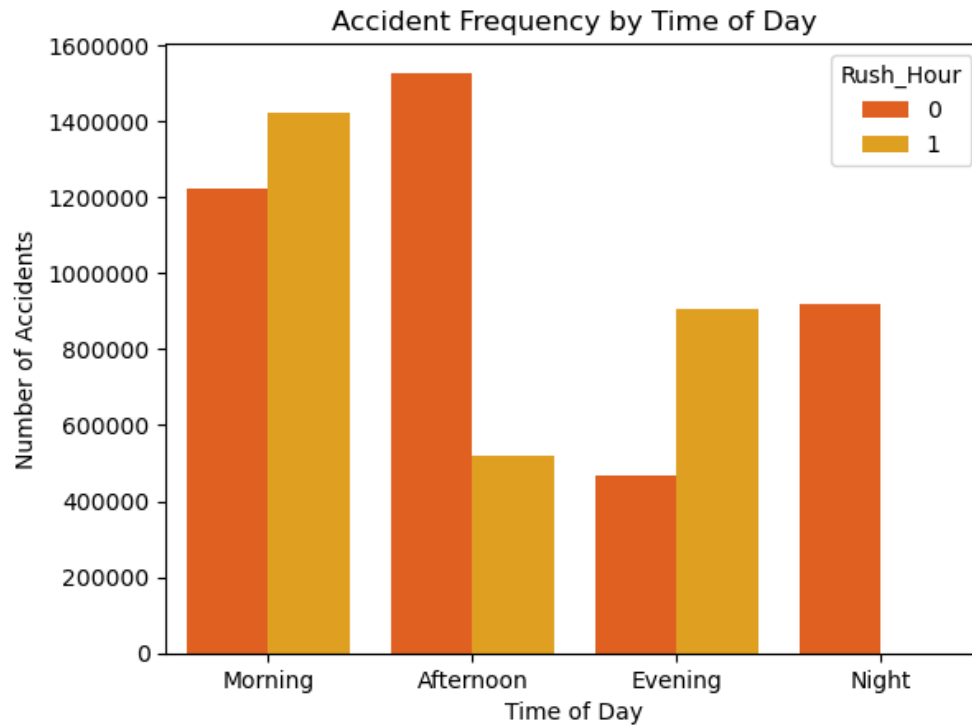


Figure 4: Accident frequency by time-of-day bucket, split by rush-hour status

Key Insights:

- Accident frequency is significantly higher during Morning and Evening rush-hour windows, reflecting heavy commuter traffic.
- Afternoon hours report the highest overall accident count outside rush hours, likely due to sustained daytime traffic from work, school, and commercial activity.
- Nighttime accidents are comparatively lower and occur mostly outside rush hours, consistent with reduced traffic volume late at night.

4.2.4 Accidents by Light Conditions

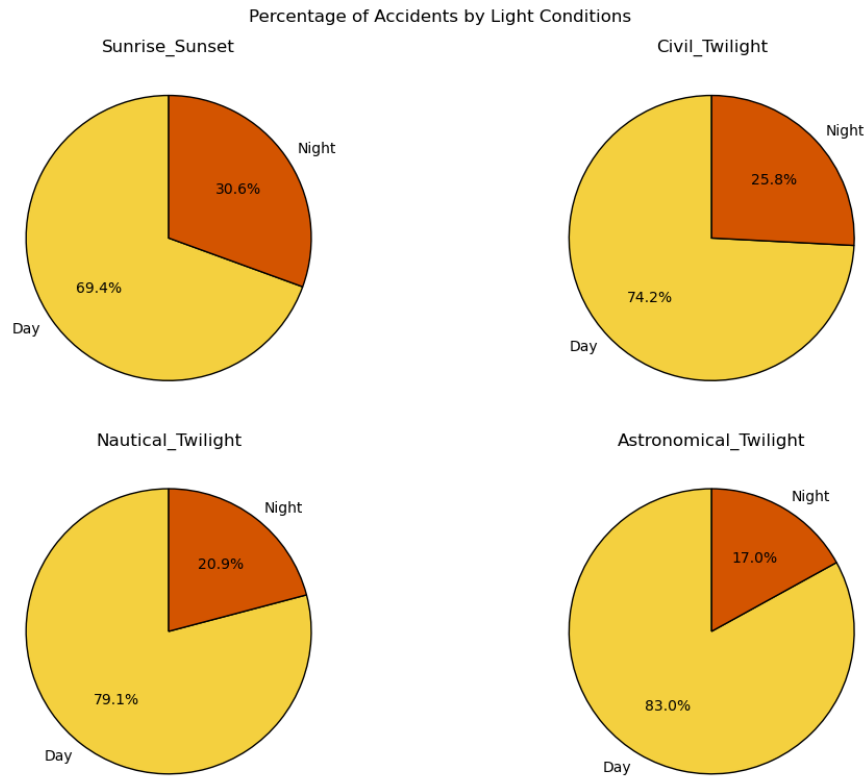


Figure 5: Share of accidents by Sunrise/Sunset and twilight condition

Key Insights:

- Most accidents occur during daylight conditions (roughly 69–83% across all twilight categories), consistent with higher traffic volume during the day.
- Nighttime accidents account for a smaller share (roughly 17–31%), reflecting fewer vehicles on the road after dark.
- As lighting conditions darken, the daytime share of accidents increases further, suggesting accident frequency is driven more by traffic density than by lighting alone.

4.2.5 Accidents by Month vs Year

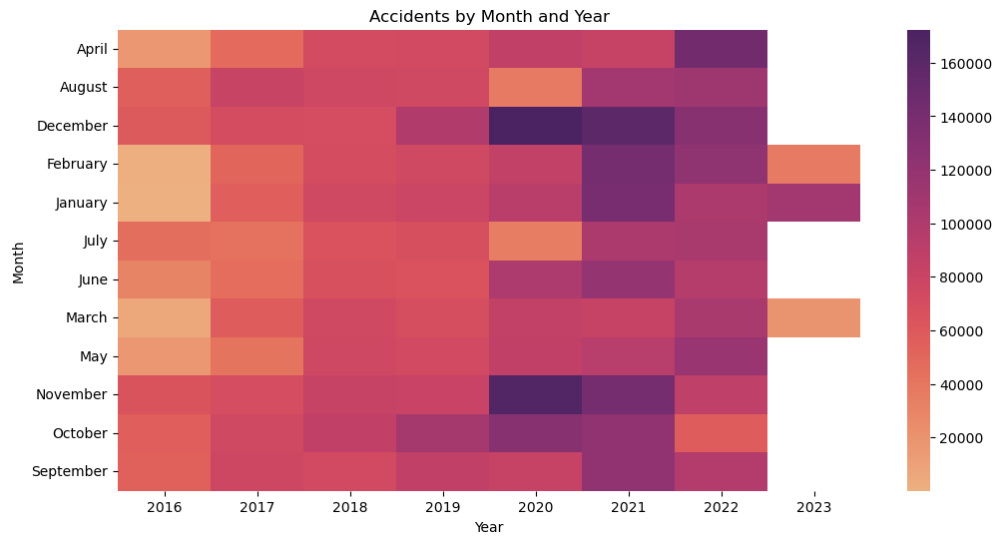


Figure 6: Heatmap of accident counts by month and year (2016–2023)

Key Insights:

- The heatmap shows how accident frequency evolved year over year from 2016 to 2023.
- From 2016 to 2018, there is a moderate increase in accidents, corresponding to summer travel and holiday seasons (May–July).
- The COVID-19 period (early 2020) shows a visible dip in accident intensity as stay-at-home orders and travel restrictions reduced road usage, followed by a rebound with notable spikes around November–December 2021 as travel resumed.

4.3 Environmental Risk Factors

4.3.1 Accidents by Weather Condition and Month

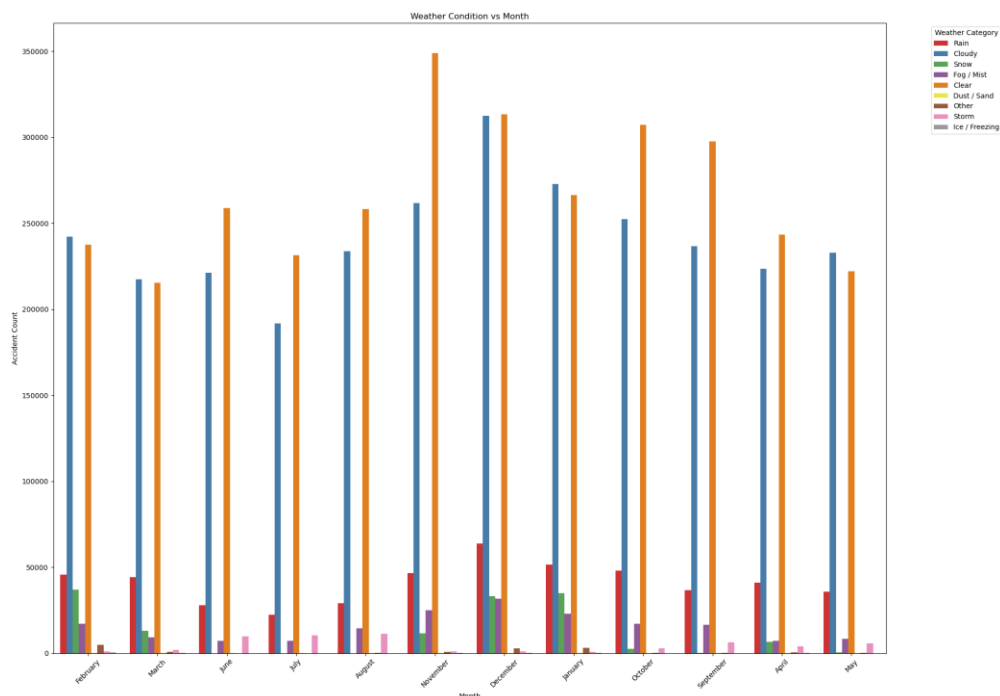


Figure 7: Accident counts by month, split by weather category

Key Insights:

- Across all months, accidents under Clear and Cloudy conditions are consistently the highest, indicating overall accident frequency is driven more by traffic volume than by adverse weather.
- Rain-related accidents remain consistently present across months, with noticeable peaks in late summer and early winter, likely due to wet-road conditions reducing tire friction.
- Snow-related accidents increase significantly during winter months, particularly December, consistent with icy/snowy road conditions.

4.3.2 Visibility vs Severity & Wind Speed vs Severity

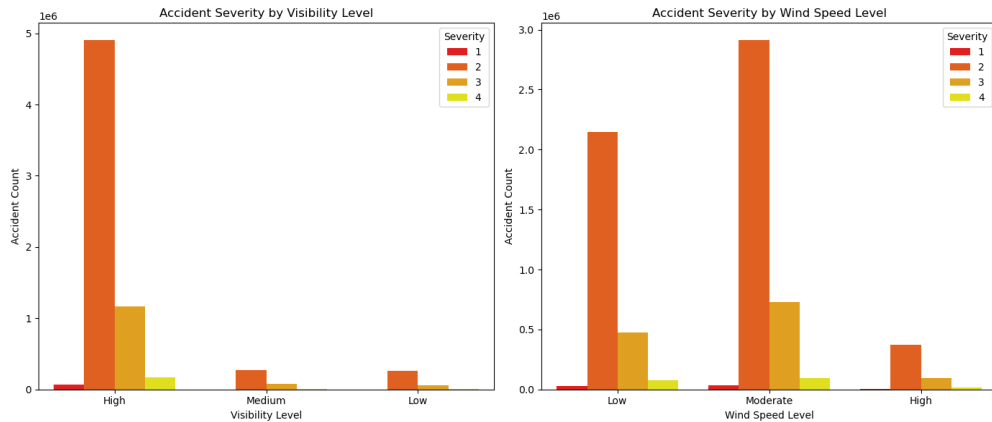


Figure 8: Accident severity by visibility level (left) and wind-speed level (right)

Key Insights:

- Most accidents happen when visibility is high, suggesting accidents are more closely tied to normal high-traffic driving conditions than to poor visibility itself.
- Severity Level 2 accidents are the most common across all visibility levels.
- Accidents are most frequent under moderate wind speeds, likely because these are the most common driving conditions overall; very strong winds and very low visibility are comparatively rare situations.

4.3.3 Accidents Near Road Infrastructure

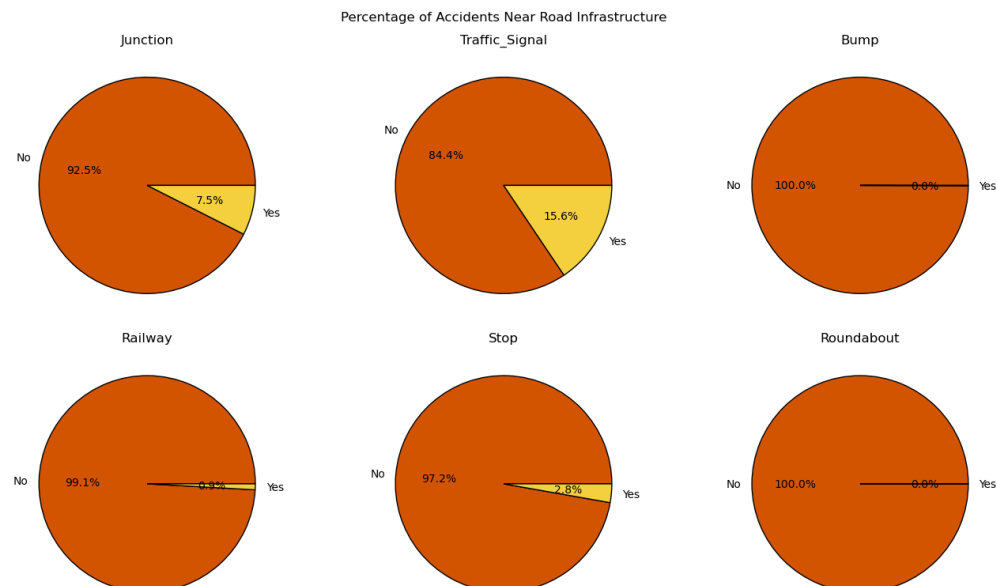


Figure 9: Share of accidents occurring near six types of road infrastructure

Key Insights:

- Only a small share of accidents occurs near junctions, most accidents happen away from junction areas.
- Traffic signals show the highest share of accidents among the infrastructure types examined, suggesting intersections with signals remain comparatively risky due to sudden stopping, turning, and congestion.
- Very few accidents are recorded near speed bumps and roundabouts, suggesting these features may be effective in reducing accident occurrence at those locations.

4.4 Geospatial Analysis

4.4.1 Top 10 States by Accident Count

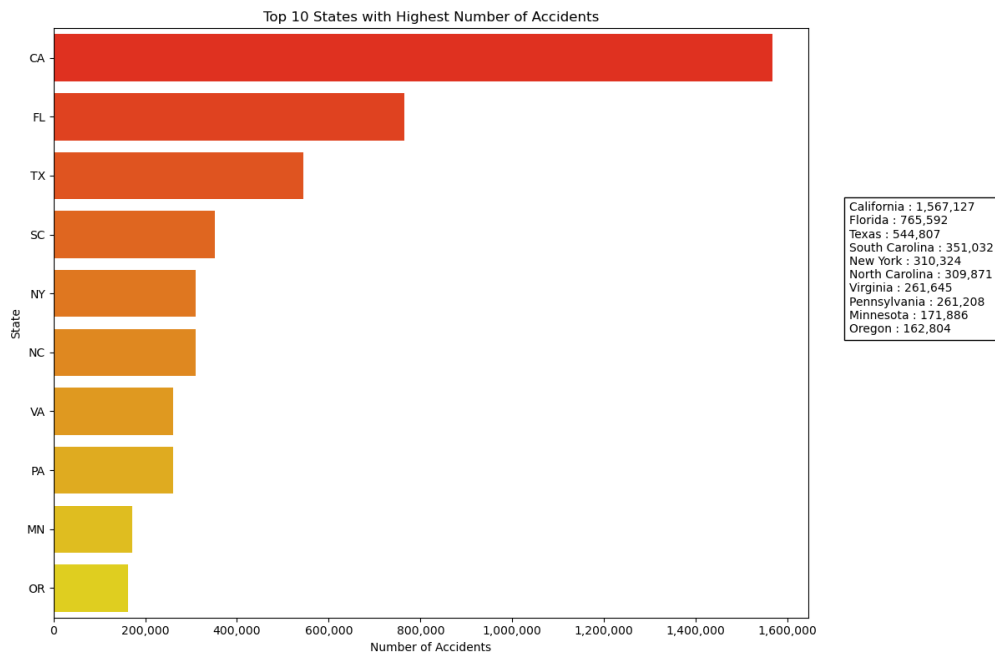


Figure 10: Top 10 US states ranked by total accident count

4.4.2 Accident Hotspot Map Across the USA

Traffic Accident Hotspots Across the USA

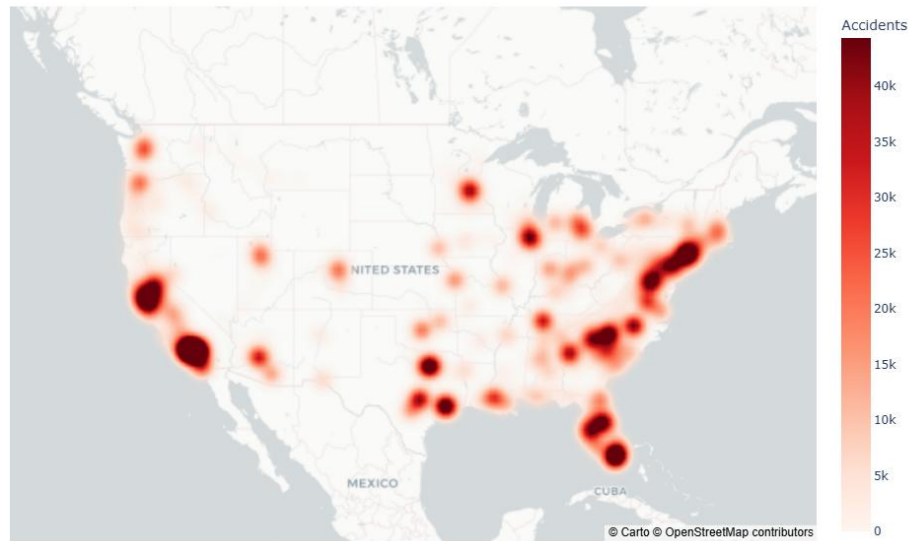


Figure 11: Density map of accident hotspots across the United States

4.5 Key Findings from the Exploratory Analysis

- Traffic accidents are heavily concentrated in major metropolitan regions, with the strongest clusters in Southern California, the Texas urban corridor, Florida, and the Northeast United States.
- Accident frequency increases significantly during rush-hour commuting periods, highlighting the influence of traffic congestion and daily mobility patterns on road safety.
- Urban infrastructure such as intersections, traffic signals, and junctions is a notable risk factor, reflecting complex vehicle interactions and higher decision-making demands at these locations.
- Weather-related factors (rain, snow, low visibility) show seasonal patterns but are secondary to traffic-volume effects in driving overall accident frequency.

5. From EDA to Machine Learning: Predicting Accident Severity Risk

The exploratory analysis repeatedly showed that severity risk is not random, it clusters around rush hours, specific weather categories, visibility levels, and road infrastructure. This section turns that observation into a trained model that estimates the probability an accident will be severe (Severity 3–4), given only the conditions that would be known at or before the time of the accident.

5.1 Problem Framing

Business question: Given the conditions of a trip at a moment in time, location, weather, time of day, and nearby road infrastructure, what is the probability that an accident, if it occurs, will be severe?

This was framed as a binary classification problem (Severe_Accident: 1 vs 0) rather than a regression on raw severity, since the operational use case (e.g. dispatch prioritization) needs a clear "high risk vs not" signal rather than a fine-grained severity score.

5.2 Leakage Audit, Choosing Features Honestly

Before model-building, columns were audited for label leakage, fields that would only be known after the fact and would let the model "cheat":

- Severity, excluded, since it is the direct source of the target variable.
- Distance(mi), excluded, since it represents how much road length was affected by the accident, which is only known after the accident occurred.
- Accident_Duration_Min, excluded for the same reason: it is only observable after the accident has ended.

The final feature set used 31 leakage-safe features: 8 numeric (Temperature, Humidity, Pressure, Visibility, Wind_Speed, Precipitation, Hour, Year), 15 binary (Weekend, Rush_Hour, and 13 road-infrastructure indicators), and 8 categorical (State, Weather_Category, Visibility_Level, Wind_Level, Time_of_Day, Season, Day_of_Week, Sunrise_Sunset).

5.3 Sampling Strategy

At close to 7 million rows, training gradient-boosted trees with cross-validation on the full dataset is slow and memory-intensive. A stratified sample of 1,000,000 rows was drawn instead, stratified on the target so the severe/non-severe ratio (21.24%) is preserved exactly in the sample, matching the full dataset's rate to within 0.01 percentage points.

5.4 Preprocessing Pipeline

A single ColumnTransformer handled all three feature types consistently: numeric features were standardized (zero mean, unit variance); binary features were passed through as-is; and categorical features were one-hot encoded with unknown categories ignored at inference time. This was wrapped in a scikit-learn Pipeline together with the model, so preprocessing was refit only on training data at each split.

5.5 Time-Based Train/Test Split

Rather than a random split, the data was split by calendar time, training on 2016–2021 and testing on 2022–2023, to mirror how the model would actually be used: only past data is available to learn from, and the model must generalize to accidents it has not seen. This resulted in a training set of 793,814 rows (24.71% severe) and a test set of 206,186 rows (7.90% severe).

5.6 Model Comparison (Cross-Validated)

Three candidate models were compared using 3-fold stratified cross-validation on a 150,000-row training sub-sample, scored by ROC-AUC:

Model	Mean ROC-AUC	Std. Dev.
Logistic Regression	0.747	± 0.001
Random Forest	0.770	± 0.002
XGBoost	0.781	± 0.001

XGBoost achieved the highest mean ROC-AUC and was selected as the final model, consistent with gradient-boosted trees generally performing well on structured/tabular data with mixed categorical and numeric features. Its more mature feature-importance and SHAP tooling was also a factor, given the project's emphasis on interpretability.

5.7 Final Model: XGBoost with Imbalance Handling

The final model was trained on the full 2016–2021 training set (not just the cross-validation sub-sample) to use as much signal as possible. Class imbalance was handled via `scale_pos_weight = 3.05` (the ratio of non-severe to severe examples in training), XGBoost's built-in mechanism for up-weighting the minority (severe) class. Key hyperparameters: 300 estimators, max depth 6, learning rate 0.1, subsample 0.8, `colsample_bytree` 0.8.

5.8 Evaluation on the 2022–2023 Holdout

Accuracy was intentionally not used as the headline metric, with roughly 1 in 13 test-set accidents being severe, a model that never predicts "severe" would still score close to 92% accuracy while being operationally useless. Precision, recall, F1, ROC-AUC, and PR-AUC were used instead, since recall on the severe class matters most for a dispatch/alerting use case.

Class	Precision	Recall	F1-score	Support
Non-Severe	0.93	0.91	0.92	189,888
Severe	0.20	0.25	0.22	16,298
Accuracy			0.86	206,186

Overall: ROC-AUC = 0.702, PR-AUC = 0.165 (against a random baseline of 0.079). The model captures meaningful signal above the class-imbalance baseline, though the severe class remains harder to identify precisely, a realistic result on this kind of imbalanced, real-world accident-severity task.

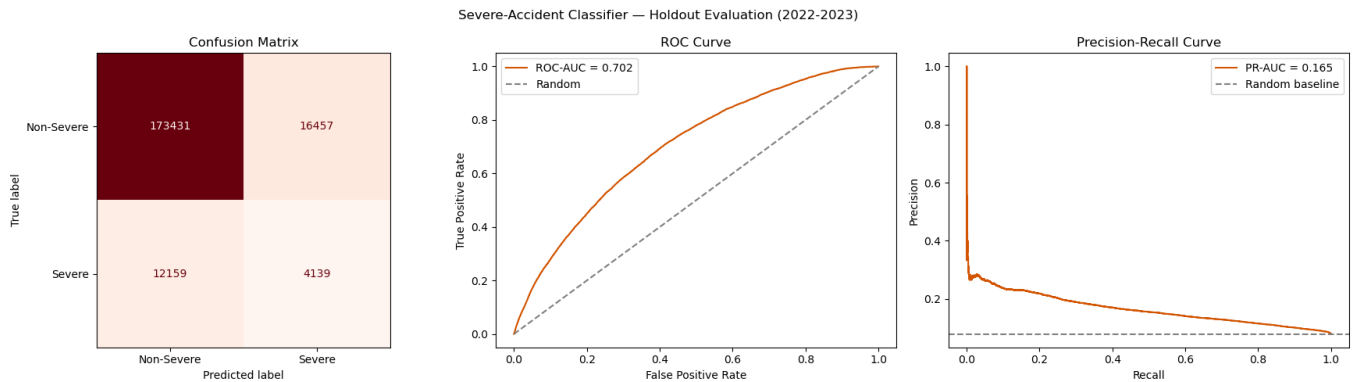


Figure 12: Confusion matrix, ROC curve, and Precision-Recall curve on the 2022–2023 holdout set

5.9 Feature Importance: Connecting Back to the EDA

Feature importance (gain-based) shows which inputs the trained XGBoost model relied on most heavily:

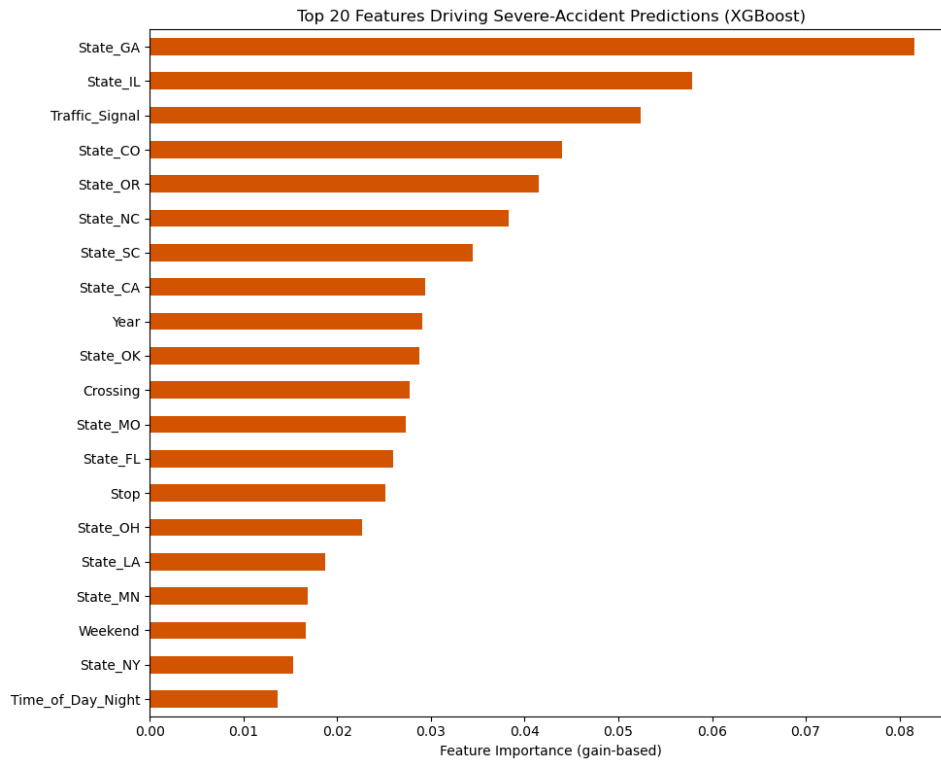


Figure 13: Top 20 features driving severe-accident predictions

Rank	Feature	Importance
1	State_GA	0.0816
2	State_IL	0.0579
3	Traffic_Signal	0.0524
4	State_CO	0.0440
5	State_OR	0.0415
6	State_NC	0.0383
7	State_SC	0.0344
8	State_CA	0.0294
9	Year	0.0291
10	State_OK	0.0288

State-level location dominates the top features, alongside Traffic_Signal and Year, indicating that where an accident happens (which implicitly captures state-level road design, speed limits, and reporting practices) is the single strongest predictor of severity, with proximity to a traffic signal and the year of the accident also contributing meaningfully. This aligns with the EDA's geographic hotspot findings in Section 4.4.

5.10 SHAP: Explaining Individual Predictions

While feature importance shows what matters on average across the whole model, SHAP (SHapley Additive exPlanations) values show, for any single prediction, how much each feature pushed that specific prediction toward "severe" or "non-severe".

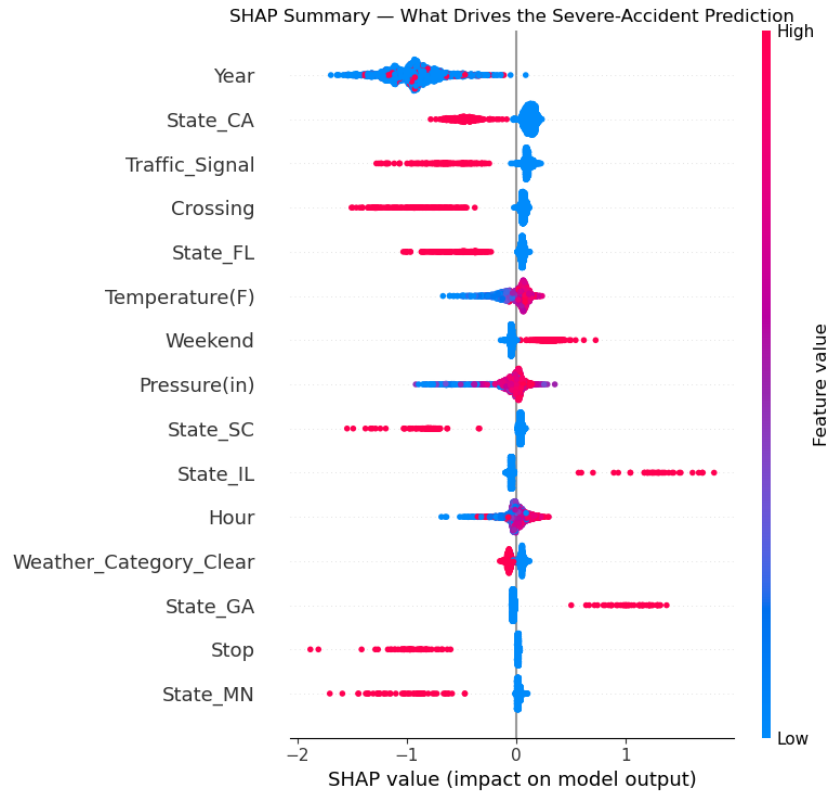


Figure 14: SHAP summary plot, feature impact on individual severe-accident predictions

Each point represents one accident from the holdout sample; red indicates a high value of that feature, blue a low value, and horizontal position shows whether that feature pushed the prediction toward severe (right) or non-severe (left) for that specific case. This provides a direct, per-prediction explanation rather than treating the model as a black box.

5.11 Bonus: Unsupervised Risk-Tier Clustering of Hotspots

As a complement to the single-density hotspot map in Section 4.4, accident locations were binned geographically (0.1-degree lat/lng bins) and clustered with K-Means on two dimensions: accident volume and severe-accident rate. This distinguishes, for example, a low-traffic location with a high severe-accident rate from a high-traffic but lower-severity urban intersection, two meaningfully different kinds of "hotspot".

Accident Hotspots Grouped by Risk Tier (Volume + Severe-Accident Rate)



Figure 15: Accident hotspots grouped into K-Means risk tiers by volume and severe-accident rate

6. Conclusion

This project analyzed 7.7 million US traffic accident records to identify the factors driving accident frequency and severity. Rush hour, weather, visibility, and road infrastructure all showed clear, consistent patterns in the exploratory analysis. Building on those findings, a classifier was trained to predict whether an accident will be severe, using only conditions that would be known at the time, with a data-leakage audit performed before training to ensure the evaluation reflects genuine predictive signal rather than information the model would not have in practice.

Key outcomes:

- Cleaned and optimized a 7.7M-row, 6 GB dataset down to 1.3 GB (≈29% reduction) while preserving all information needed for analysis.
- Engineered 12 new features connecting raw timestamps and weather readings to interpretable categories used throughout the analysis.
- Identified rush hour, urban infrastructure, and geographic concentration in major metro regions as the dominant patterns in accident frequency.
- Audited for and removed data leakage (Severity, Distance, Accident_Duration_Min) before model training.
- Compared three models with cross-validation and selected XGBoost (ROC-AUC 0.781 in cross-validation) as the final model.
- Achieved ROC-AUC 0.702 and PR-AUC 0.165 (vs. a 0.079 random baseline) on a genuinely out-of-time 2022–2023 holdout set.
- Explained the model’s behavior at both the global level (feature importance) and the individual-prediction level (SHAP), tying the model’s learned signal back to the original EDA findings.
- Added an unsupervised risk-tier clustering layer on top of the hotspot map, distinguishing high-volume and high-severity-rate locations.

Together, these results move from a purely descriptive analysis of what happened in the past to a model capable of estimating severity risk for a given set of conditions going forward, with the geographic and infrastructural drivers identified in the EDA independently confirmed by the model's own feature importances.